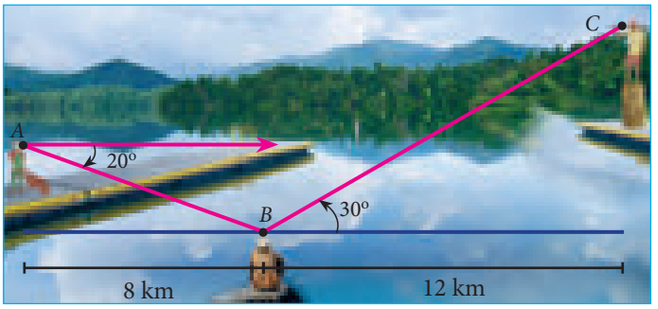
2. A man is standing on the deck of a ship, which is 40 m above water level. He observes the angle of elevation of the top of a hill as $60°$ and the angle of depression of the base of the hill as $30°$. Calculate the distance of the hill from the ship and the height of the hill. $(\sqrt{3} = 1.732)$

3. If the angle of elevation of a cloud from a point '$h$' metres above a lake is $\theta_1$ and the angle of depression of its reflection in the lake is $\theta_2$. Prove that the height that the cloud is located from the ground is $\frac{h(\tan\theta_1 + \tan\theta_2)}{\tan\theta_2 - \tan\theta_1}$.

4. The angle of elevation of the top of a cell phone tower from the foot of a high apartment is $60°$ and the angle of depression of the foot of the tower from the top of the apartment is $30°$. If the height of the apartment is $50$ m, find the height of the cell phone tower. According to radiations control norms, the minimum height of a cell phone tower should be $120$ m. State if the height of the above mentioned cell phone tower meets the radiation norms.

5. The angles of elevation and depression of the top and bottom of a lamp post from the top of a 66 m high apartment are $60°$ and $30°$ respectively. Find
   (i) The height of the lamp post.
   (ii) The difference between height of the lamp post and the apartment.
   (iii) The distance between the lamp post and the apartment. $(\sqrt{3} = 1.732)$

6. Three villagers $A$, $B$ and $C$ can see each other using telescope across a valley. The horizontal distance between $A$ and $B$ is 8 km and the horizontal distance between $B$ and $C$ is 12 km. The angle of depression of B from $A$ is $20°$ and the angle of elevation of $C$ from $B$ is $30°$. Calculate : (i) the vertical height between $A$ and $B$. (ii) the vertical height between $B$ and $C$. $(\tan 20° = 0.3640,\ \sqrt{3} = 1.732)$




## Exercise 6.5

### Multiple choice questions

1. The value of $\sin^2\theta + \frac{1}{1+\tan^2\theta}$ is equal to

   (A) $\tan^2\theta$ (B) 1 (C) $\cot^2\theta$ (D) 0

2. $\tan\theta\operatorname{cosec}^2\theta - \tan\theta$ is equal to

   (A) $\sec\theta$ (B) $\cot^2\theta$ (C) $\sin\theta$ (D) $\cot\theta$

3. If $(\sin\alpha + \operatorname{cosec}\alpha)^2 + (\cos\alpha + \sec\alpha)^2 = k + \tan^2\alpha + \cot^2\alpha$, then the value of $k$ is equal to

   (A) 9 (B) 7 (C) 5 (D) 3



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